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A survey of trust and reputation systems for online service provision

Audun Jøsang [a], Roslan Ismail [b], Colin Boyd [b]

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Abstract

Trust and reputation systems represent a significant trend in decision support for Internet mediated service provision. The basic idea is to let parties rate each other, for example after the completion of a transaction, and use the aggregated ratings about a given party to derive a trust or reputation score, which can assist other parties in deciding whether or not to transact with that party in the future. A natural side effect is that it also provides an incentive for good behaviour, and therefore tends to have a positive effect on market quality. Reputation systems can be called collaborative sanctioning systems to reflect their collaborative nature, and are related to collaborative filtering systems. Reputation systems are already being used in successful commercial online applications. There is also a rapidly growing literature around trust and reputation systems, but unfortunately this activity is not very coherent. The purpose of this article is to give an overview of existing and proposed systems that can be used to derive measures of trust and reputation for Internet transactions, to analyse the current trends and developments in this area, and to propose a research agenda for trust and reputation systems.

Introduction

Online service provision commonly takes place between parties who have never transacted with each other before, in an environment where the service consumer often has insufficient information about the service provider, and about the goods and services offered. This forces the consumer to accept the “risk of prior performance”, i.e. to pay for services and goods before receiving them, which can leave him in a vulnerable position. The consumer generally has no opportunity to see and try products, i.e. to “squeeze the oranges”, before he buys. The service provider, on the other hand, knows exactly what he gets, as long as he is paid in money. The inefficiencies resulting from this information asymmetry can be mitigated through trust and reputation. The idea is that even if the consumer cannot try the product or service in advance, he can be confident that it will be what he expects as long as he trusts the seller. A trusted seller therefore has a significant advantage in case the product quality cannot be verified in advance.

This example shows that trust plays a crucial role in computer mediated transactions and processes. However, it is often hard to assess the trustworthiness of remote entities, because computerised communication media are increasingly removing us from familiar styles of interaction. Physical encounter and traditional forms of communication allow people to assess a much wider range of cues related to trustworthiness than is currently possible through computer mediated communication. The time and investment it takes to establish a traditional brick-and-mortar street presence provides some assurance that those who do it are serious players. This stands in sharp contrast to the relative simplicity and low cost of establishing a good looking Internet presence which gives little evidence about the solidity of the organisation behind it. The difficulty of collecting evidence about unknown transaction partners makes it hard to distinguish between high and low quality service providers on the Internet. As a result, the topic of trust in open computer networks is receiving considerable attention in the academic community and e-commerce industry.

There is a rapidly growing literature on the theory and applications of trust and reputation systems, and the main purpose of this document is to provide a survey of the developments in this area. An earlier brief survey of reputation systems has been published by Mui et al. [50]. Overviews of agent transaction systems are also relevant because they often relate to reputation systems [25], [42], [38]. There is considerable confusion around the terminology used to describe these systems, and we will try to describe proposals and developments using a consistent terminology in this study. There also seems to be a lack of coherence in this area, as indicated by the fact that authors often propose new systems from scratch, without trying to extend and enhance previous proposals.

Section 2 attempts to define the concepts of trust and reputation, and proposes an agenda for research into trust and reputation systems. Section 3 describes why trust and reputation systems should be regarded as security mechanisms. Section 4 describes the relationship between collaborative filtering systems and reputation systems, where the latter can also be defined in terms of collaborative sanctioning systems. In Section 5 we describe different trust classes, of which *provision trust* is a class of trust that refers to service provision. Section 6 describes four categories for reputation and trust semantics that can be used in trust and reputation systems, Section 7 describes centralised and computation methods, i.e. how ratings are to be computed to derive reputation scores. Section 9 provides an overview of reputation systems in commercial and live applications. Section 10 describes the main problems in reputation systems, and provides an overview of literature that proposes solutions to these problems. The study is rounded off with a discussion in Section 11.

The notion of trust

Manifestations of trust are easy to recognise because we experience and rely on it everyday, but at the same time trust is quite challenging to define because it manifests itself in many different forms. The literature on trust can also be quite confusing because the term is being used with a variety of meanings [46]. Two common definitions of trust which we will call *reliability trust* and *decision trust* respectively will be used in this study.

As the name suggest, reliability trust can be …

Trust and reputation systems as soft security mechanisms

In a general sense, the purpose of security mechanisms is to provide protection against malicious parties. In this sense there is a whole range of security challenges that are not met by traditional approaches. Traditional security mechanisms will typically protect resources from malicious users, by restricting access to only authorised users. However, in many situations we have to protect ourselves from those who offer resources so that the problem in fact is reversed. Information providers …

Collaborative filtering and collaborative sanctioning

Collaborative filtering systems (CF) have similarities with reputation systems in that both collect ratings from members in a community. However they also have fundamental differences. The assumptions behind CF systems is that different people have different tastes, and rate things differently according to subjective taste. If two users rate a set of items similarly, they share similar tastes, and are called *neighbours* in the jargon. This information can be used to recommend items that one …

Trust classes

In order to be more specific about trust semantics, we will distinguish between a set of different trust classes according to Grandison and Sloman's classification [23]. This is illustrated in Fig. 2.[3] The highlighting of provision trust in Fig. 2 is done to illustrate that it is the focus of the trust and reputation systems described in this study.

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Categories of trust semantics

The semantic characteristics of ratings, reputation scores and trust measures are important in order for participants to be able to interpret those measures. The semantics of measures can be described in terms of a *specificity-generality* dimension and a *subjectivity-objectivity* dimension as illustrated in Table 1.

A specific measure means that it relates to a specific trust aspect such as the ability to deliver on time, whereas a general measure is supposed to represent an average of all aspects. …

Reputation network architectures

The technical principles for building reputation systems are described in this and the following section. The network architecture determines how ratings and reputation scores are communicated between participants in a reputation system. The two main types are centralised and distributed architectures. …

Reputation computation engines

Seen from the relying party's point of view, trust and reputation scores can be computed based on own experience, on second hand referrals, or on a combination of both. In the jargon of economic theory, the term *private information* is used to describe first hand information resulting from own experience, and *public information* is used to describe publicly available second hand information, i.e. information that can be obtained from third parties.

Reputation systems are typically based on public …

Commercial and live reputation systems

This section describes the most well known applications of online reputation systems. All analysed systems have a centralised network architecture. The computation is mostly based on the summation or average of ratings, but two systems use the flow model. …

Problems and proposed solutions

Numerous problems exist in all practical and academic reputation systems. This section describes problems that have been identified and some proposed solutions. …

Discussion and conclusion

The purpose of this work has been to describe and analyse the state of the art in trust and reputation systems. Dingledine et al. [16] have proposed the following set of basic criteria for judging the quality and soundness of reputation computation engines.

(1) *Accuracy for long-term performance*. The system must reflect the confidence of a given score. It must also have the capability to distinguish between a new entity of unknown quality and an entity with poor long-term performance. …